

Bacterial DNA Isolation, Purification & Quantification

Laboratory Report — Template DNA Preparation for PCR

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Technique category: Molecular Biology

1. Objective

To isolate genomic DNA from a bacterial culture, purify it to remove protein and RNA contamination, and assess its concentration and purity by spectrophotometry to determine suitability as PCR template DNA.

2. Introduction

Bacterial DNA extraction involves three core stages: cell lysis to release DNA from the bacterial cell wall and membrane, removal of protein, RNA, and cellular debris, and precipitation and washing of the purified DNA. Because PCR is highly sensitive to inhibitors such as residual protein, phenol, or ethanol, extracted DNA must be assessed for both concentration and purity before use. Purity is commonly assessed by the A260/A280 absorbance ratio: a ratio close to 1.8 indicates high-purity DNA, while deviations suggest protein or RNA contamination.

3. Materials

- Overnight bacterial culture (broth or plate colony)
- Lysis buffer with lysozyme (for gram-positive strains)
- 10% SDS, Proteinase K, RNase A
- Phenol-chloroform-isoamyl alcohol (25:24:1) or a commercial extraction kit
- Isopropanol and 70% ethanol
- TE buffer or nuclease-free water
- NanoDrop spectrophotometer
- Agarose gel electrophoresis apparatus (integrity check)

4. Method

- **Lysis:** Bacterial pellet resuspended in lysis buffer (+ lysozyme for gram-positive strains); SDS and Proteinase K added, incubated at 55–60°C.
- **Purification:** RNase A treatment, followed by phenol-chloroform extraction; aqueous (DNA-containing) phase collected.
- **Precipitation:** DNA precipitated with cold isopropanol, pelleted by centrifugation, washed with 70% ethanol, and resuspended in TE buffer.
- **Quantification:** DNA concentration and A260/A280 purity ratio measured by spectrophotometry.
- **Quality check:** Aliquot run on agarose gel to confirm a single high-molecular-weight band with minimal smearing.

5. Results

Five extracted DNA samples were assessed for concentration and A260/A280 purity ratio. Samples with a ratio between 1.75 and 1.90 were classified as suitable for PCR; ratios below 1.75 indicated likely protein contamination, and ratios above 1.90 indicated possible RNA contamination.

Sample	Conc. (ng/uL)	A260/A280	Status
Sample_1	145.2	1.83	Suitable for PCR
Sample_2	88.6	1.52	Protein contamination
Sample_3	210.4	2.15	Possible RNA contamination
Sample_4	62.1	1.80	Suitable for PCR
Sample_5	175.8	1.80	Suitable for PCR

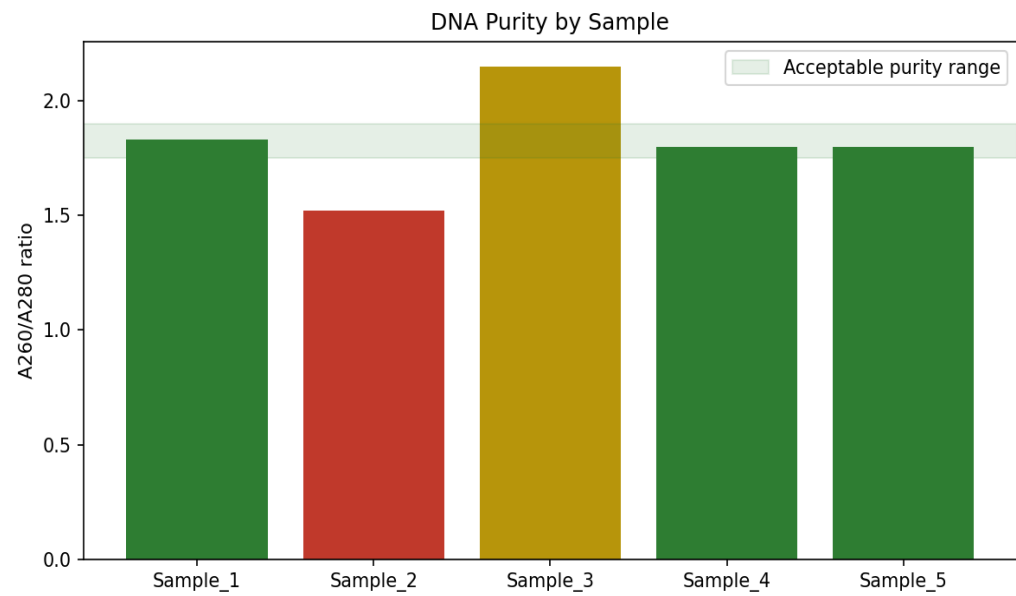


Figure 1. A260/A280 purity ratio per sample, with the acceptable purity range (1.75–1.90) shaded.

Three of the five samples (Sample_1, Sample_4, Sample_5) fell within the acceptable purity range and were confirmed suitable for use as PCR template. Sample_2 showed a low ratio consistent with residual protein contamination, and Sample_3 showed an elevated ratio consistent with residual RNA contamination.

6. Discussion

The spread of purity ratios across the five samples illustrates common failure points in DNA extraction: Sample_2's low ratio suggests incomplete phenol-chloroform separation or insufficient Proteinase K digestion, leaving residual protein bound to the DNA. Sample_3's elevated ratio suggests RNase A treatment was incomplete or insufficient, leaving co-purified RNA. Both would be expected to reduce PCR efficiency if used directly as template — Sample_2 through possible polymerase inhibition by residual protein, and Sample_3 through reduced effective DNA concentration and potential well-to-well competition in a NanoDrop-only volume estimate. In practice, both samples would be re-purified (repeat phenol-chloroform extraction or ethanol wash) before use in PCR, and the three samples that met the acceptable range would be diluted to a standard working concentration (commonly 10–50 ng/uL) with nuclease-free water prior to amplification.

7. Conclusion

This exercise demonstrated the full bacterial DNA extraction and quality-control workflow, from lysis through spectrophotometric purity assessment, and highlighted why purity checks — not concentration alone — determine whether extracted DNA is fit for use as PCR template.